STAT 333 Notes

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1 Review of Elementary Probability

1.1 Probability Space

Experiment

A fair coin is tossed once

Sample Space

Set of all possible outcomes (Ω), $\Omega \in \{H, T\}$

Events

Roughly set of all subsets of sample space Ω, $\epsilon = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$

1.2 Probability

Probability is a function from the set of all events ϵ to $\mathbb{R}$, $P : \epsilon \to \mathbb{R}$

- for any $A \in \epsilon$, $P(A) \in [0, 1]$
- $P(\Omega) = 1$
- for any $n \in \mathbb{Z}^+$, $E_1, E_2, \cdots, E_n \in \epsilon$ such that $E_i \cap E_j = \emptyset$ for $i \neq j$, then $P(\cup_{i=1}^{n} E_i) = \sum_{i=1}^{n} P(E_i)$

Properties

- $P(\emptyset) = 0$
- $A, B \in \epsilon$, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- for any $A \in \epsilon$, $P(A^c) = 1 - P(A)$
- if A, B satisfies $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$
- If $A \subseteq B$, then $P(A) \leq P(B)$

1.3 Conditional Probability

The conditional probability of an event A given B is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

and is only defined when $P(B) > 0$

Properties

- $P(A \cap B) = P(A|B)P(B)$
- for any $A_1, A_2, \cdots, A_n$, $P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \cdots P(A_n|A_1 \cap A_2 \cap \cdots \cap A_{n-1})$
- $P(A^c|B) = 1 - P(A|B)$
- A_1, A_2 are two events, $P(A_1 \cup A_2|B) = P(A_1|B) + P(A_2|B) - P(A_1 \cap A_2|B)$

1.4 Independent Events

Two events A and B are mutually independent of each other if $P(A \cap B) = P(A)P(B)$

The events $A_1, A_2, \cdots, A_n$ are mutually independent if $P(A_1 \cap A_2 \cap \cdots \cap A_n) = \prod_{i=1}^{n} P(A_i)$

- if A and B are independent of each other, $P(A|B) = P(A)$

1.5 Law of Total Probability

Say $E_1, \cdots, E_n$ are disjoint events. If satisfies $E_i \cap E_j = \emptyset$ for $i \neq j$, $\cup_{i=1}^{n} E_i = \Omega$. Then for any event A,

$$P(A) = \sum_{i=1}^{n} P(A|E_i)P(E_i)$$

1.6 Bayes' Formula

$$\begin{aligned} P(E_i|A) &= \frac{P(A \cap E_i)}{P(A)} \\ &= \frac{P(A \cap E_i)}{\sum_{j=1}^{n} P(A \cap E_j)} \\ &= \frac{P(A|E_i)P(E_i)}{\sum_{j=1}^{n} P(A|E_j)P(E_j)} \end{aligned}$$

1.7 Random Variables

A function from sample space to real numbers $X : \omega \to \mathbb{R}$. Range of X: state space

1.8 Discrete Random Variables

If state space / range of random variable is discrete ($\{a_1, a_2, \cdots\}$)

Probability Mass Function (PMF)

$$P(a) = P(X = a)$$

Cumulative Distribution Function (CDF)

$$F(a) = P(X \leq a) = \sum_{x \leq a} P(X = x)$$

Tail Probability Function (TPF)

$$\overline{F}(a) = P(X > a) = 1 - F(a)$$

Properties

State space $= \{a_1, a_2, \cdots\}$, $a_1 < a_2 < \cdots$

$$p(a_i) = F(a_i) - F(a_{i-1})$$

1.9 Discrete Distributions

Bernoulli Distribution

$x \in \{0, 1\}$, $X \sim Ber(p)$

$$p(x) = p^x (1-p)^{1-x}$$

Binomial Distribution

x: number of successes in n independent $Ber(p)$ trials, $x \in \{0, 1, \cdots, n\}$, $X \sim Bin(n, p)$

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

Geometric Distribution

$x_1, x_2, \cdots, \overset{iid}{\sim} Ber(p)$
y: number of trials required to get the first success, $y \in \{1, 2, \cdots\}$, $Y \sim Geo(p)$

$$p_Y(x) = P(Y = x) = (1-p)^{x-1} p$$

z: number of failures until the first success, $z \in \{0, 1, \cdots\}$, $Z \sim Geo(p)$

$$p_Z(x) = P(Z = x) = (1-p)^x p$$

Poisson Distribution

$X \sim Poi(\lambda)$, $x \in \{0, 1, \cdots\}$, $\lambda > 0$

$$p(x) = \frac{\lambda^x}{x!} e^{-\lambda}$$

1.10 Continuous Random Variables

x is continuous random variable, x takes values over continuous in $\mathbb{R}$

Probability Density Function (PDF)

$f_X : \mathbb{R} \to [0, \infty)$, $\displaystyle\int_{-\infty}^{\infty} f_X(x)dx = 1$, $F(x) = P(X \leq x) = \displaystyle\int_{-\infty}^{x} f_X(t)dt$, $f_X(x) = \dfrac{dF(x)}{dx}$

1. For continuous x, cdf $F(x)$ is continuous
2. For any interval $B \subseteq \mathbb{R}$
$$P(x \in B) = \int_B f_X(x)dx$$
, if $B \in (-\infty, x] \to P(x \in B) = F(x)$

Uniform Distribution

$X \sim U(a, b)$, $a < b$

$$f_X(x) = \begin{cases} \dfrac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Exponential Distribution

$X \sim Exp(\lambda)$, $\lambda > 0$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Erlang Distribution

$X \sim Erlang(n, \lambda)$, $n \in \{1, 2, \cdots\}$, $\lambda > 0$

$$f_X(x) = \begin{cases} \dfrac{\lambda^n}{(n-1)!} x^{n-1} e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Gamma Distribution

$X \sim Gamma(\alpha, \lambda)$, $\alpha > 0$, $\lambda > 0$

$$f_X(x) = \begin{cases} \dfrac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

1.11 Expectation

X is a continuous random variable, $g : \mathbb{R} \to \mathbb{R}$ is a function

$$E[g(X)] = \begin{cases} \displaystyle\int_{-\infty}^{\infty} g(x) f_X(x) dx & \text{if } g(x) \text{ is continuous} \\ \displaystyle\sum_{x \in \mathbb{R}} g(x) p_X(x) & \text{if } g(x) \text{ is discrete} \end{cases}$$

Some special $g(x)$

1. $g(x) = x^n \Rightarrow E[g(X)] = E[X^n]$, n^{th} moment of X
2. $\mu = E(X)$, $g(x) = (x - \mu)^2 \Rightarrow E[g(X)] = E[(X - \mu)^2] = Var(X)$
3. $Var(X) = E(X^2) - [E(X)]^2$
4. $E(aX + b) = aE(X) + b$, $a, b \in \mathbb{R}$
5. $Var(aX + b) = a^2 Var(X)$, $a, b \in \mathbb{R}$

1.12 Moment Generating Function (MGF)

Properties

- $M_X(t) = E[e^{tX}]$, $t \in \mathbb{R}$
- $\left.\dfrac{d^n}{dt} M_X(t)\right|_{t=0} = E[X^n]$
- x, y are two random variables and identically distributed iff $M_X(t) = M_Y(t)$
- $x_1, x_2, \cdots, x_n$ are independent, then

$$M_{X_1+X_2+\cdots+X_n}(t) = M_{X_1}(t) M_{X_2}(t) \cdots M_{X_n}(t) = \prod_{i=1}^{n} M_{X_i}(t)$$

1.13 Joint Distribution

Two random variables X and Y defined on the same sample space
cdf / Joint cdf:

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

Marginal cdf:

$$F_X(x) = P(X \leq x) = \lim_{y \to \infty} F_{X,Y}(x, y), \;\; F_Y(y) = P(Y \leq y) = \lim_{x \to \infty} F_{X,Y}(x, y)$$

1.13.1 Discrete Joint Distribution

Joint pmf: $p_{X,Y}(x,y) = P(X = x, Y = y)$

$$P_X(x) = \sum_y p_{X,Y}(x,y),\ P_Y(y) = \sum_x p_{X,Y}(x,y)$$

1.13.2 Continuous Joint Distribution

Joint pdf: $f_{X,Y}(x,y) = \dfrac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y)$

$$F_{X,Y}(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(t,s)dtds$$

1.14 Expectation

$$E[g(X,Y)] = \begin{cases} \displaystyle\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f_{X,Y}(x,y)dydx & \text{if } g(x,y) \text{ is continuous} \\ \displaystyle\sum_{x \in \mathbb{R}} \sum_{y \in \mathbb{R}} g(x,y) p_{X,Y}(x,y) & \text{if } g(x,y) \text{ is discrete} \end{cases}$$

$E[g(X,Y)]$ exists if one of the following happens

- $E[g(X,Y)] < \infty$
- $g(X,Y) \geq 0$
- $g(X,Y) \leq 0$

1.15 Independence of Random Variables

X and Y are independent if $F_{X,Y}(x,y) = F_X(x)F_Y(y)$
For discrete case:

$$P_{X,Y}(x,y) = P_X(x)P_Y(y)$$

For continuous case:

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

For expectation, if X and Y are independent, then

$$E[g(X,Y)] = E[g(X)]E[g(Y)]$$

1.16 Covariance

$$\begin{aligned} Cov(X,Y) &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E[XY] - E[X]E[Y] \end{aligned}$$

1.16.1 Properties

- If X and Y are independent, then $Cov(X, Y) = 0$, but $Cov(X, Y) = 0$ does not imply X and Y are independent
- X and Y are independent, then $M_{X,Y}(t, s) = M_X(t)M_Y(s)$
- X and Y are independent, then $Var(aX + bY) = a^2Var(X) + b^2Var(Y)$

1.17 Convergence Modes

$X_1, X_2, \cdots$ is a sequence of random variables, X is another random variable

1. $X_n \to X$ almost surely $P(X_n \to X) = 1$. $\omega \in \Omega$ (one element of sample space), $X_n(\omega) \to X(\omega)$ as $n \to \infty$. $E = \{\omega \in \Omega : X_n(\omega) \to X(\omega)\}$, $P(E) = 1$
2. $X_n \to X$ in probability if for any $\epsilon > 0$, $P(|X_n - X| > \epsilon) \to 0$ as $n \to \infty$
3. $X_n \to X$ in distribution if $F_{X_n}(a) \to F_X(a)$ for every a where $F_X(a)$ is continuous

1.17.1 Property

$X_n \to X$ almost surely $\Rightarrow X_n \to X$ in probability $\Rightarrow X_n \to X$ in distribution
Not the other way around

1.18 Strong Law of Large Numbers (SLLN)

$X_1, X_2, \cdots$ are a sequence of iid random variables with $E(|X_1|) < \infty$, then

$$\frac{X_1 + \cdots + X_n}{n} \to E(X_1) \text{ almost surely as } n \to \infty$$

2 Conditional Distributions and Conditional Expectation

2.1 Conditional Probability

Discrete Case

X_1, X_2 are discrete random variables, the conditional pmf of X_1 given X_2 is

$$P_{X_1|X_2}(x_1|x_2) = P(X_1 = x_1|X_2 = x_2) = \frac{P(X_1 = x_1, X_2 = x_2)}{P(X_2 = x_2)}$$

is defined only when $P(X_2 = x_2) > 0$

Properties

If $P_{X_1}(x_1), P_{X_2}(x_2) > 0$,

$$P_{X_2|X_1}(x_2|x_1) = \frac{P_{X_1|X_2}(x_1|x_2)P_{X_2(x_2)}}{P(X_1 = x_1)}$$

If X_1, X_2, X_3 are discrete and $P_{X_3}(x_3) > 0$, then

$$P_{X_1,X_2|X_3}(x_1, x_2|x_3) = \frac{P_{X_1,X_2,X_3}(x_1, x_2, x_3)}{P_{X_3}(x_3)}$$

Continuous Case

X and Y are two continuous random variables,

$$f_{X,Y}(x, y) : \text{joint pdf of } X, Y$$

$$f_X(x), f_Y(y) : \text{marginal pdf of } X, Y$$

The conditional pdf of $X|Y = y$ is

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

is defined only when $f_Y(y) > 0$

Properties

Assume that both $f_X(x)$ and $f_Y(y)$ are positive, then

1. $f_{Y|X}(y|x) = \dfrac{f_{X|Y}(x|y)f_Y(y)}{f_X(x)}$
2. $f_X(x) = \displaystyle\int f_{X|Y}(x|y)f_Y(y)dy$
3. $f_Y(y) = \displaystyle\int f_{Y|X}(y|x)f_X(x)dx$
4. $f_{Y|X}(y|x) = \dfrac{f_{X|Y}(x|y)f_Y(y)}{\int f_{X|Y}(x|y)f_Y(y)dy}$

2.2 Conditional Expectation

Discrete Case

$$E[X_1|X_2 = x_2] = \sum_{x_1} x_1 P_{X_1|X_2}(x_1|x_2)$$

More generally, if $w(\cdot,\cdot)$, $h(\cdot)$, and $g(\cdot)$ are functions, then

$$E(w(X_1, X_2)|X_2 = x_2) = E(w(X_1, x_2)|X_2 = x_2) = \sum_{x_1} w(x_1, x_2) P_{X_1|X_2}(x_1|x_2)$$

and

$$E(g(X_1)h(X_2)|X_2 = x_2) = E[g(X_1)h(x_2)|X_2 = x_2] = h(x_2)E[g(X_1)|X_2 = x_2]$$
$$E[ag(X_1) + bh(X_1)|X_2 = x_2] = aE[g(X_1)|X_2 = x_2] + bE[h(X_1)|X_2 = x_2]$$

Properties

If $X_1, X_2, \cdots, X_n, Y$ jointly discrete random variables, and $P_Y(y) > 0$, and for any $a_1, \cdots, a_n \in \mathbb{R}$, then

$$E[a_1X_1 + a_2X_2 + \cdots + a_nX_n|Y = y] = \sum_{i=1}^{n} a_i E[X_i|Y = y]$$

Continuous Case

$E[g(Y)|X = x] = \int g(y) f_{Y|X}(y|x)dy$ Special case:

- $g(y) = y$, then $E[Y|X = x] = \int y f_{Y|X}(y|x)dy$
- $g(y) = (Y - E[Y|X = x])^2$, then $E[(Y - E[Y|X = x])^2|X = x] = Var(Y|X = x)$

2.3 Independence

If X_1, X_2 are independent, then

- $P_{X_1|X_2}(x_1|x_2) = P_{X_1}(x_1)$
- $E[X_1|X_2] = E[X_1]$

2.4 Mix of Discrete and Continuous Random Variables

X is continuous random variable, and Y is discrete random variable
Situation 1: $f_X(x)$ and $P_{Y|X}(y|x)$ are given, then

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{P_Y(y)} = \frac{f_X(x)P_{Y|X}(y|x)}{P_Y(y)}$$

Situation 2: $P_Y(y)$ and $f_{X|Y}(x|y)$ are given, then

$$P_{Y|X}(y|x) = \frac{f_{X|Y}(x|y)P_Y(y)}{f_X(x)}$$

2.5 Compute Expectation by Conditioning

X and Y are two random variables, and $g : \mathbb{R} \to \mathbb{R}$, then

$$E[g(X)] = E[E[g(X)|Y]]$$

Thoerem

If X and Y are two random variables, then

$$Var(X) = E[Var(X|Y)] + Var(E[X|Y])$$

2.6 Calculating Probabilities using Conditioning

Suppose we have two continuous random variables X and Y

$$\begin{aligned}
P(X < Y) &= \int P(X < Y|Y = y) f_Y(y) dy \\
&= \int P(X < y|Y = y) f_Y(y) dy \\
&\text{if } X \text{ and } Y \text{ are independent} \\
&= \int F_X(y) f_Y(y) dy \\
&= E[F_X(Y)]
\end{aligned}$$

If X and Y are identically distributed, then

$$\begin{aligned}
P(X < Y) &= \int P(X < Y|Y = y) f_Y(y) dy \\
&= \int P(Y > X|Y = y) f_Y(y) dy \\
&= \int (1 - F_X(y)) f_Y(y) dy \\
&= 1 - E[F_X(Y)] \\
&= 1/2
\end{aligned}$$

3 Discrete-Time Markov Chains

3.1 Definitions and Basic Concepts

3.1.1 Stochastic Process

$\{X(t), t \in T\}$ is called a stochastic process if $X(t)$ is a rv for any given $t \in T$. T is referred to as the index set and is often interpreted in the context of time

3.1.2 Discrete-Time Markov Chain (DTMC), Markov Property

A stochastic process $\{X_n, n \in \mathbb{N}\}$ is said to be a discrete-time Markov chain if the following conditions hold true:

- For $n \in \mathbb{N}$, X_n is discrete rv
- For $n \in \mathbb{N}$ and all states $x_0, x_1, \cdots, x_{n+1} \in S$, the Markov property must hold:

  $$P(X_{n+1} = x_{n+1} | X_n = x_n, X_{n-1} = x_{n-1}, \cdots, X_0 = x_0) = P(X_{n+1} = x_{n+1} | X_n = x_n)$$

  The current state X_n contains all the information needed to predict the next state X_{n+1}, regardless of the previous states $X_{n-1}, \cdots, X_0$

3.1.3 Transition Probability

For any pair of states i and j, the transition probability from state i at time n to state j at time $n+1$ is given by

$$P_{n,i,j} = P(X_{n+1} = j | X_n = i)$$

3.1.4 Stationary Transition Probability

A DTMC is called stationary or homogeneous if the transition probabilities do not depend on time

$$P(X_n = j | X_{n-1} = i) = P(X_{n-1} = j | X_{n-2} = i) = \cdots = P(X_1 = j | X_0 = i) = P_{i,j}$$

3.1.5 Transition Probability Matrix

One Step Transition Probability Matrix

Let P_n be the associated matrix containing all these transition probabilities, referred to as the one-step transition probability matrix (TPM) from time n to time $n+1$. It looks like

$$P_n = [P_{n,i,j}] = \begin{bmatrix} P_{n,0,0} & P_{n,0,1} & \cdots & P_{n,0,j} \\ P_{n,1,0} & P_{n,1,1} & \cdots & P_{n,1,j} \\ \vdots & \vdots & \ddots & \vdots \\ P_{n,i,0} & P_{n,i,1} & \cdots & P_{n,i,j} \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix}$$

Stationary and Homogeneous

For each pair of states i and j, if $P_{n,i,j}$ does not depend on n, $P_{n,i,j} = P_{i,j}$, $\forall n \in \mathbb{N}$, then we say the DTMC is stationary or homogeneous, one-step TPM is

$$P = [P_{i,j}] = \begin{bmatrix} P_{0,0} & P_{0,1} & \cdots & P_{0,j} \\ P_{1,0} & P_{1,1} & \cdots & P_{1,j} \\ \vdots & \vdots & \ddots & \vdots \\ P_{i,0} & P_{i,1} & \cdots & P_{i,j} \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix}$$

$$\begin{aligned} P_{n,i,j} &= P(X_{n+1} = j | X_n = i) \\ &= P(X_1 = j | X_0 = i) \\ &= P_{i,j} \end{aligned}$$

N-Step Transition Probability

For any pair of states i and j, the n-step transition probability is given by

$$P_{i,j}^{(n)} = P(X_n = j | X_0 = i) = P(X_{m+n} = j | X_m = i) \quad m, n \in \mathbb{N}$$

Due to stationary assumption, this quantity is actually independent of m. Thus $P_{i,j}^{(n)} = P(X_n = j | X_0 = i), n \in \mathbb{N}$

$$P_{i,j}^{(n)} = P(X_0 = j | X_0 = i) = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

3.1.6 Properties of Stationary Transition Matrix

Say $P = [P_{i,j}]_{i,j \in S}$ is stationary transition matrix, then

- $P_{i,j} \geq 0$
- $\sum_j P_{i,j} = 1$

3.1.7 Chapman-Kolmogorov Equation

For $n \in \mathbb{Z}^+$

$$\begin{aligned}
P_{ij}^{(n)} &= P(X_n = j | X_0 = i) \\
&= \sum_{k=0}^{\infty} P(X_n = j | X_{n-1} = k, X_0 = i) P(X_{n-1} = k | X_0 = i) \\
&= \sum_{k=0}^{\infty} P_{i,k}^{(n-1)} P(X_n = j | X_{n-1} = k, X_0 = i) \\
&= \sum_{k=0}^{\infty} P_{i,k}^{(n-1)} P(X_n = j | X_{n-1} = k) \\
&= \sum_{k=0}^{\infty} P_{i,k}^{(n-1)} P_{k,j}
\end{aligned}$$

More generality, for $m, n \in \mathbb{Z}^+$,

$$P_{i,j}^{(m+n)} = \sum_{k \in S} P_{ik}^{(m)} P_{kj}^{(n)}$$

Property

$$P^{(n)} = P \cdot P^{(n-1)} = P^n$$

3.1.8 Initial Distribution & n-Step Distribution

α_0: the vector consisting of all the pmf of X_0, the initial distribution of DTMC
α_n: the vector consisting of all the pmf of X_n, the n-step distribution of DTMC

$$\alpha_n = \alpha_0 \cdot P^{(n)} = \alpha_0 \cdot P^n$$

3.1.9 Accessible, Communicate

State j is accessible from state i (denote by $i \to j$) if $\exists n \in \mathbb{N}$ such that $P_{i,j}^{(n)} > 0$. If states i and j are accessible from each other, then states i and j communicate (denote by $i \leftrightarrow j$), $i \leftrightarrow j$ iff $\exists m, n \in \mathbb{N}$ such that $P_{i,j}^{(n)} > 0$ and $P_{j,i}^{(m)} > 0$.

3.1.10 Equivalence Relation

The concept of communication forms what is known as an equivalence relation, satisfying the following

- Reflexivity: $i \leftrightarrow i$
- Symmetry: $i \leftrightarrow j \Rightarrow j \leftrightarrow i$
- Transitivity: $i \leftrightarrow j$ and $j \leftrightarrow k \Rightarrow i \leftrightarrow k$

3.1.11 Communication Class

Within the same communication class, all states communicate with each other
If states i and j belongs to different communication classes, then $i \leftrightarrow j$ is not true, at most one of $i \to j$ or $j \to i$ can be true

3.1.12 Reducible, Irreducible

A DTMC that has only one communication class is said to be irreducible. On the other hand, a DTMC is called reducible if there are two or more communication classes

3.1.13 Period

The period of state i is given by $d(i) = \gcd\{n \in \mathbb{Z}^+ : P_{i,i}^{(n)} > 0\}$

Aperiodic

If $d(i) = 1$, then state i is said to be aperiodic. In fact, DTMC is said to be aperiodic if $d(i) = 1$ for all $n \in \mathbb{N}$. Furthermore, if $P_{i,i}^{(n)} = 0$ for all $n \in \mathbb{Z}^+$, then we set $d(i) = \infty$

3.1.14 Theorem

If $i \leftrightarrow j$, then $d(i) = d(j)$

3.2 Transience and Recurrence

The first visit of the DTMC to state j occurs at time $n \in \mathbb{Z}^+$

$$\begin{aligned}
f_{i,j}^{(n)} &= P(\text{DTMC makes a future visit to state } j \text{ at time } n | X_0 = i) \\
&= P(X_n = j, X_{n-1} \neq j, X_{n-2} \neq j, \cdots, X_1 \neq j | X_0 = i) \\
f_{ii}^{(n)} &= P(X_n = i, X_{n-1} \neq i, X_{n-2} \neq i, \cdots, X_1 \neq i | X_0 = i) \\
f_{ii}^{(0)} &= 0
\end{aligned}$$

Result:

$$f_{ij}^{(n)} = P_{ij}^{(n)} - \sum_{k=1}^{n-1} f_{ij}^{(k)} P_{jj}^{(n-k)}$$

Definition

$$f_{ij} = P(\text{DTMC ever makes a visit to state } j | X_0 = i)$$

$$T_i^{(n)} : \text{time for DRMC to makes n-th visit at state } i$$

$$f_{i,j}^{(n)} = P(T_j^{(n)} = n | X_0 = i)$$

3.2.1 Transient, Recurrent

State i is said to be transient if $f_{i,i} < 1$. State i is said to be recurrent if $f_{i,i} = 1$

Definition

M_i: number of times a DTMC make a visit to state i

$$A_n^{(i)} = \begin{cases} 1 & \text{if } X_n = i \\ 0 & \text{if } X_n \neq i \end{cases}$$

$$M_i = \sum_{n=0}^{\infty} A_n^{(i)}$$

Given $X_0 = i$,

- if state i is recurrent, then $M_i = \infty$ with probability 1
- if state i is transient, then $M_i \sim Geo_f(1 - f_{ii})$

Definition

$T_i^{(n)}$: time at n-th arrival to state i

$$\begin{aligned} I_i^{(n)} &= T_i^{(n)} - T_i^{(n-1)} \\ &= \text{time between the } (n-1)^{th} \text{ and } n^{th} \text{ visit to state } i \end{aligned}$$

Define $T_i^{(0)} = 0$
Then $T_i^{(1)} = I_i^{(1)}$, $T_i^{(n)} = I_i^{(1)} + I_i^{(2)} + \cdots + I_i^{(n)}$

$$f_{ii} = P(T_i^{(1)} < \infty | X_0 = i) = P(I_i^{(1)} < \infty | X_0 = i)$$

Theorem

For a DTMC with stationary transition probability, the following holds

- given $X_0 = i$, the inter arrival time $I_1^{(1)}, I_1^{(2)}, \cdots$ are iid integer valued random variables
- if state i is transient, then $M_i \sim Geo_f(1 - f_{ii})$, where M_i is number of arrivals at state i
- if state i is recurrent, then $M_i = \infty$ with probability 1

Corollary

$$M_i^{(n)} = \begin{cases} 1 & \text{if } X_n = i \\ 0 & \text{if } X_n \neq i \end{cases}$$
$$= \mathbb{1}\{X_n = i\}$$

$$M_i = \sum_{n=1}^{\infty} \mathbb{1}\{X_n = i\}$$

Then

$$\begin{aligned} E[M_i|X_0 = i] &= E\left[\sum_{n=1}^{\infty} \mathbb{1}\{X_n = i\}|X_0 = i\right] \\ &= \sum_{n=1}^{\infty} E\left[\mathbb{1}\{X_n = i\}|X_0 = i\right] \\ &= \sum_{n=1}^{\infty} P(X_n = i|X_0 = i) \\ &= \sum_{n=1}^{\infty} P_{i,i}^{(n)} \end{aligned}$$

- If state i is transient
$$E[M_i|X_0 = i] = \sum_{n=1}^{\infty} P_{i,i}^{(n)} < \infty$$
- If state i is recurrent
$$E[M_i|X_0 = i] = \sum_{n=1}^{\infty} P_{i,i}^{(n)} = \infty$$
- A state is recurrent iff
$$\sum_{n=1}^{\infty} P_{i,i}^{(n)} = \infty$$
- Symmetric random walk on $\mathbb{Z}$ is recurrent, but asymmetric random walk are transient

Theorem

If $i \leftrightarrow j$ and state i is recurrent, then state j is also recurrent
If $i \leftrightarrow j$ and state i is transient, then state j is also transient

Theorem

If $i \leftrightarrow j$ and state i is recurrent, then

$$f_{i,j} = P(\text{DTMC ever makes a future visit to state } j|X_0 = i) = 1$$

Finite DTMC

DTMC is called finite if the total number of states is finite

Theorem

Consider a finite DTMC

- At least one of the states is recurrent
- If the chain is irreducible, then the DTMC is recurrent, means all states of the DTMC are recurrent
- If the chain is not irreducible, then it has at least one recurrence class

A finite-state DTMC has at least one recurrent state
An irreducible, finite-state DTMC must be recurrent

Theorem

If state i is recurrent and state i does not communicate with state j, then $P_{i,j}^{(k)} = 0$ for all $k \in \mathbb{Z}^+$

Theorem

If $i \to j$ and i is recurrent, then

- $j \to i$
- $i \leftrightarrow j$ and j is recurrent
- $f_{i,j} = P(X_n = j \text{ for some } n \geq 1 | X_0 = i) = 1,\ f_{j,i} = 1$

3.3 Limiting Behaviour of DTMCs

Theorem

For any state i and transient state j of a DTMC, $\lim_{n\to\infty} P_{i,j}^{(n)} = 0$

Definition

Let

$$N_i = \min\{n \in \mathbb{Z}^+ : X_n = i\}$$

where state i is assumed to be recurrent. The first return time to state i is defined as

$$T_i^{(1)} = N_i$$

The conditional pmf is given by

$$P(N_i = n | X_0 = i) = f_{i,i}^{(n)}, n = 1, 2, 3, \cdots$$

3.3.1 Mean Recurrent Time

If state i is recurrent, then its mean recurrent time is given by

$$m_i = E[N_i|X_0 = i] = \sum_{n=1}^{\infty} n f_{i,i}^{(n)}$$

In words, m_i represents the average time it takes the DTMC to make successive visits to state i.

3.3.2 Positive Recurrent, Null Recurrent

Suppose that state i is recurrent. State i is said to be positive recurrent if $m_i < \infty$. On the other hand, state i is said to be null recurrent if $m_i = \infty$

Theorem

- If $i \leftrightarrow j$ and state i is positive recurrent, then state j is also positive recurrent
- If $i \leftrightarrow j$ and state i is null recurrent, then state j is also null recurrent
- In a finite state DTMC, all recurrent states are positive recurrent
- In a finite state DTMC, at least one state is positive recurrent

3.3.3 Stationary Distribution

A probability distribution $\{p_i\}_{i=0}^{\infty}$ is called a stationary distribution of a DTMC if $\{p_i\}_{i=0}^{\infty}$ satisfies the conditions $\sum_{i=0}^{\infty} p_i = 1$ and $p_j = \sum_{i=0}^{\infty} p_i P_{i,j}$ for all $j \in \mathbb{N}$

$$\pi P = \pi$$

3.3.4 Convex Combination

For any 2 vectors $\pi_1 = (\pi_{11}, \pi_{12}, \cdots, \pi_{1n})$ and $\pi_2 = (\pi_{21}, \pi_{22}, \cdots, \pi_{2n})$, a convex combination of π_1 and π_2 is given by

$$\alpha\pi_1 + (1-\alpha)\pi_2 \quad 0 \leq \alpha \leq 1$$

Result

For any 2 stationary distributions π_1 and π_2, the convex combination $\alpha\pi_1 + (1-\alpha)\pi_2$ where $0 \leq \alpha \leq 1$ is another stationary distribution

$$(\alpha\pi_1 + (1-\alpha)\pi_2)P = \alpha\pi_1 + (1-\alpha)\pi_2$$

Theorem

$T_x^{(1)}$: time to first return to state x

$$m_x = E[T_x^{(1)} | X_0 = x]$$

$$f_{xy} = P(X_n = y, n \geq 1 | X_0 = x)$$

$$\frac{1}{n} \sum_{n=1}^{n} P_{x,y}^{(n)} \to \frac{f_{xy}}{m_y} \quad n \to \infty$$

- $m_y = \infty$ if y is transient or null recurrent
- $m_y < \infty$ if y is positive recurrent

Theorem

If the DTMC is

- irreducible
- aperiodic
- positive recurrent

Then the stationary distribution is unique and

$$\lim_{n \to \infty} P_{x,y}^{(n)} = \pi_y = \frac{1}{m_y}$$

$$\lim_{n \to \infty} P(X_n = x) = \pi_x = \frac{1}{m_x}$$

For a positive recurrent state y, the expected return time is

$$m_y = \frac{1}{\pi_y}$$

Result

- Stationary distribution exists iff at least one of the state is positive recurrent
  - For finite DTMC, at least one state is positive recurrent, hence stationary distribution always exists
- If more than one positive recurrent classes, then stationary distribution is not unique
- For finite chain the stationary distribution is unique if it has only one positive recurrent class

3.3.5 Basic Limit Theorem

For an irreducible, recurrent, and aperiodic DTMC, $\lim_{n\to\infty} P_{i,j}^{(n)}$ exists and is independent of state i, satisfying

$$\lim_{n\to\infty} P_{i,j}^{(n)} = \pi_j = \frac{1}{m_j} \quad \forall i, j \in \mathbb{N}$$

If the DTMC also happens to be positive recurrent, then $\{\pi_j\}_{j=0}^{\infty}$ is the unique, positive solution to the system of linear equations defined by

$$\begin{cases} \pi_j = \sum_{i=0}^{\infty} \pi_i P_{i,j} & \forall j \in \mathbb{N} \\ \sum_{j=0}^{\infty} \pi_j = 1 \end{cases}$$

Theorem

Suppose that a finite-state DTMC with state space $S = \{0, 1, \cdots, N-1\}$ is irreducible and aperiodic. If the associated TPM is doubly stochastic, then the limiting probabilities $\{\pi_j\}_{j=0}^{N-1}$ exist and are given by

$$\pi_j = \frac{1}{N} \quad j = 0, 1, \cdots, N-1$$

3.3.6 Probability Generating Function

$$\begin{aligned} \alpha(t) &= \sum_{j=0}^{\infty} t^j \alpha_j \\ &= E[t^{Z_{0,1}}] \end{aligned}$$

$$\left.\frac{d^n}{dt^n}(\alpha(t))\right|_{t=0} = (n!) \cdot \alpha_n$$

Let $\pi_0 = P(X_n = 0 \text{ for some } n \geq 1 | X_0 = 1)$

- 1 is a solution to $\pi_0 = \alpha(\pi_0)$
- π_0 is a solution to the equation
$$t = \alpha(t)$$
- $f(t) = \alpha(t) - t$ is a convex function in $[0, 1]$
- Since $f(t) = \alpha(t) - t$ is strictly convex, smooth and non constant inside $[0, 1]$, the equation $f(t) = 0$ has at most two solution in $[0, 1]$

Theorem

$\pi_0 = \lim_{n\to\infty} P(X_n = 0)$ is the smallest solution of $t = \alpha(t)$ within the interval $[0, 1]$

4 The Exponential Distribution and the Poisson Process

4.1 Properties of the Exponential Distribution

If a rv X has an exponential distribution with parameter $\lambda > 0$ ($X \sim Exp(\lambda)$ where λ is often referred to as the "rate"), then we have the following distributional results

- pdf: $f(x) = \lambda e^{-\lambda x}, x \geq 0$
- cdf: $F(x) = 1 - e^{-\lambda x}, x \geq 0$
- tpf: $\overline{F}(x) = e^{-\lambda x}, x \geq 0$
- mgf: $\phi_X(t) = \frac{\lambda}{\lambda - t}, t < \lambda$
- mean: $E[X] = \frac{1}{\lambda}$
- variance: $Var(X) = \frac{1}{\lambda^2}$

Minimum of Independent Exponential

Let $\{X_i\}_{i=1}^n$ be a sequence of independent rvs where $X_i \sim Exp(\lambda_i)$, $i = 1, 2, \cdots, n$. Define $Y = \min\{X_1, X_2, \cdots, X_n\}$ to be the smallest order statistic of $\{X_1, \cdots, X_n\}$. Then

$$Y = \min\{X_1, X_2, \cdots, X_n\} \sim Exp(\sum_{i=1}^n \lambda_i)$$

Usuful Result

Let $\{X_i\}_{i=1}^n$ be a sequence of independent rvs where $X_i \sim Exp(\lambda_i)$, $i = 1, 2, \cdots, n$

- For $j \in \{1, 2, \cdots, n\}$

$$P(X_j = \min\{X_1, X_2, \cdots, X_n\}) = \frac{\lambda_j}{\sum_{i=1}^n \lambda_i}$$

- $$P(X_1 < X_2 < \cdots < X_n) = \frac{\prod_{i=1}^{n-1} \lambda_i}{\prod_{i=1}^{n-1} \left(\sum_{j=i}^n \lambda_j\right)}$$

4.1.1 Memoryless Property

A rv X is memoryless iff

$$P(X > y + z | X > y) = P(X > z) \quad \forall y, z \geq 0$$

X is memoryless iff

$$P(X > y + z) = P(X > y)P(X > z) \quad \forall y, z \geq 0$$

Theorem

An exponential distribution is memoryless

Theorem

If $X_1 \sim Exp(\lambda_1)$ and $X_2 \sim Exp(\lambda_2)$ are independent rvs, then

$$P(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2}$$

Useful Results

If $X \sim Exp(\lambda)$

$$P(X > Y + Z | X > Y) = P(X > Z)$$

If $X \sim Exp(\lambda)$, then

$$P(X - Y > z | X > Y) = P(X > z)$$

$$(X - Y) | (X > Y) \sim Exp(\lambda)$$

4.2 The Poisson Process

A counting process $\{N(t), t \geq 0\}$ is a stochastic process in which $N(t)$ represents the number of events that happen (or occur) by time t, where the index t measures time over a continuous range

Basic Properties of Counting Processes

- $N(0) = 0$
- $N(t)$ is a non-negative integer $\forall t \geq 0$ ($N(t) \in \mathbb{N} \quad \forall t \geq 0$)
- If $s < t$, then $N(s) \leq N(t)$
- $N(t) - N(s)$ counts the number of events to occur in the time interval $(s, t]$ for $s < t$

4.2.1 Independent Increments

A counting process $\{N(t), t \geq 0\}$ has independent increments if $N(t_1) - N(s_1)$ is independent of $N(t_2) - N(s_2)$ whenever $(s_1, t_1] \cap (s_2, t_2] = \emptyset$ for all choices of s_1, t_1, s_2, t_2

4.2.2 Stationary Increments

A counting process $\{N(t), t \geq 0\}$ has stationary increments if the distribution of the number of events in $(s, s+t]$ depends only on t, the length of the time interval. $N(s+t) - N(s)$ has the same probability distribution as $N(t) - N(0) = N(t)$, the number of events occurring in the interval $[0, t]$

4.2.3 Of order h

A function $y = f(x)$ is said to be "$o(h)$" (of order h) if

$$\lim_{h \to 0} \frac{f(x)}{h} = 0$$

4.2.4 Poisson Process

A counting process $\{N(t), t \geq 0\}$ is said to be a Poisson process at rate λ if the following three conditions hold true

- The process has both independent and stationary increments
- For $h > 0$, $P(N(h) = 1) = \lambda h + o(h)$
- For $h > 0$, $P(N(h) \geq 2) = o(h)$

Theorem

If $\{N(t), t \geq 0\}$ is a Poisson process at rate λ, then

$$N(t) \sim Poi(\lambda t)$$

4.2.5 Interarrival Times

Define T_1 to be the elapsed time (from time 0) until the first event occurs. In general, for $i \geq 2$, let T_i be the elapsed time between the occurences of the $(i-1)^{th}$ and i^{th} events. The sequence $\{T_i\}_{i=1}^{\infty}$ is called the interarrival or interevent time sequence

Theorem

If $\{N(t), t \geq 0\}$ is a Poisson process at rate $\lambda > 0$, then $\{T_i\}_{i=1}^{\infty}$ is a sequence of i.i.d. $Exp(\lambda)$ rvs

$$S_n = \sum_{i=1}^{n} T_i \sim Erlang(n, \lambda)$$

$$P(S_n > t) = e^{-\lambda t} \sum_{j=0}^{n-1} \frac{(\lambda t)^j}{j!}, t \geq 0$$

4.3 Further Properties of the Poisson Process

Theorem

If $\{N(t), t \geq 0\}$ is a Poisson process at rate λ, then

$$N(s)|(N(t) = n) \sim Bin(n, s/t)$$

where $s < t$

4.3.1 Comparison of Event Occurences

Suppose that $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ are independent Poisson process at rate λ_1 and λ_2 respectively. Let $S_m^{(1)}$ be the arrival time of the m-th event for $\{N_1(t), t \geq 0\}$ and $S_n^{(2)}$ be the arrival time of the n-th event for $\{N_2(t), t \geq 0\}$. We know that $S_m^{(1)} = \sum_{i=1}^m T_i^{(1)}$ where $\{T_i^{(1)}\}_{i=1}^\infty$ is a sequence of iid $Exp(\lambda_1)$ rvs and $S_n^{(2)} = \sum_{j=1}^n T_j^{(2)}$ where $\{T_j^{(2)}\}_{j=1}^\infty$ is a sequence of iid $Exp(\lambda_2)$ rvs. Then the following holds true
The probability of observing m success occur before n failure in a sequence of independent Bernoulli trials

$$P(S_m^{(1)} < S_n^{(2)}) = \sum_{j=0}^{n-1} \binom{m+j-1}{m-1} \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^m \left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^j$$

Theorem

Suppose that $\{N(t), t \geq 0\}$ is a Poisson process at rate λ. Given $N(t) = 1$, the conditional distributino of the first arrival time is uniformly distributed on $(0, t)$

$$S_1|(N(t) = 1) \sim U(0, t)$$

4.3.2 Order Statistics

The sequence of rvs $\{Y_{(1)}, Y_{(2)}, \cdots, Y_{(n)}\}$ are called the order statistics of $\{Y_1, Y_2, \cdots, Y_n\}$, satisfying

$$\begin{aligned}
Y_{(1)} &\equiv \text{1st smallest among } \{Y_1, Y_2, \cdots, Y_n\} \\
Y_{(2)} &\equiv \text{2nd smallest among } \{Y_1, Y_2, \cdots, Y_n\} \\
&\vdots \\
Y_{(n)} &\equiv \text{nth smallest among } \{Y_1, Y_2, \cdots, Y_n\}
\end{aligned}$$

Theorem

Let $\{N(t), t \geq 0\}$ be a Poisson process at rate λ. Given $N(t) = n$, the conditional joint distribution of the n arrival times is identical to the joint distribution of the n order statistics from the $U(0, t)$ distributino

$$(S_1, S_2, \cdots, S_n)|(N(t) = n) \sim (Y_{(1)}, Y_{(2)}, \cdots, Y_{(n)})$$

where $\{Y_i\}_{i=1}^n$ is an i.i.d. sequence of $U(0, t)$ rvs

4.4 Two Important Generalizations

4.4.1 The Non-Homogeneous Poisson Process

The counting process $\{N(t), t \geq 0\}$ is a non-homogeneous (non-stationary) Poisson process with rate function $\lambda(t)$ if the following three conditions hold true

- $\{N(t), t \geq 0\}$ has independent increments
- $P(N(t+h) - N(t) = 1) = h\lambda(t) + o(h)$
- $P(N(t+h) - N(t) \geq 2) = o(h)$

Theorem

If $\{N(t), t \geq 0\}$ is a non-homogeneous Poisson process with rate functionn $\lambda(t)$, then $N(s_1 + s_2) - N(s_1) \sim Poi(m(s_1 + s_2) - m(s_1))$ where the mean value function $m(t)$ is given by

$$m(t) = \int_0^t \lambda(\tau) d\tau, \quad t \geq 0$$

4.4.2 The Compound Poisson Process

Let $\{Y_i\}_{i=1}^{\infty}$ be an i.i.d. sequence of rvs. Let $\{N(t), t \geq 0\}$ be a Poisson process at rate λ, independent of each Y_i, $i = 1, 2, \cdots$. If $X(t) = \sum_{i=1}^{N(t)} Y_i$, then the process $\{X(t), t \geq 0\}$ is a compound Poisson process

$$E[X(t)] = \lambda t E(Y_1)$$
$$Var(X(t)) = \lambda t E(Y_1^2)$$

5 Useful References

5.1 Ratio Test

For a sequence of positive number $\{a_n\}$

- If $L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n} < 1$, then $\sum_{n=1}^{\infty} a_n < \infty$
- If $L > 1$, then $\sum_{n=1}^{\infty} a_n = \infty$
- If $L = 1$, the test is inconclusive

5.2 Stirling's Approximation

$$n! \sim \sqrt{2\pi} e^{-n} n^{n+\frac{1}{2}}$$